In [ ]:
```
import pandas as pd
dataset=pd.read_csv("insurance.csv")
dataset
dataset=pd.get_dummies(dataset,drop_first=True)
dataset
```

In [ ]:
```
x=dataset[['age', 'bmi', 'children','sex_male', 'smoker_yes','region_northwest','region_southeast','region_southwest']]
y=dataset['charges']
print(x)
print(y)
```

In [3]:
```
from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.20,random_state=42)
```

In [4]:
```
from sklearn.preprocessing import StandardScaler
sc=StandardScaler()
x_train=sc.fit_transform(x_train)
x_test=sc.transform(x_test)
```

In [12]:
```
from sklearn.ensemble import GradientBoostingRegressor
reg=GradientBoostingRegressor(n_estimators=500,
    max_depth=4,
    min_samples_split=5,
    learning_rate=0.01,
    loss="squared_error")
reg.fit(x_train,y_train)
```

Out[12]:

▾ GradientBoostingRegressor

▸Parameters

▸Fitted attributes

In [13]:
```
y_pred=reg.predict(x_test)
y_pred
```

```
from sklearn.metrics import r2_score
result=r2_score(y_pred,y_test)
result
```

Out[13]:

```
0.8589121129980305
```

In [ ]: